

How much performance do agricultural machine-learning models lose out of distribution? A systematic review and quantitative synthesis

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Abstract

Crop-production models are usually tested on data from the fields, seasons and sensors they were trained on, then used elsewhere. We measured the performance such models lose under out-of-distribution (OOD) evaluation. Following a pre-specified protocol and PRISMA 2020, we searched OpenAlex and Semantic Scholar (1–2 October 2026) for studies reporting in-distribution (ID) and OOD performance of the same model on a disease, weed, plant-organ, trait, crop-mapping or yield task. Of 7,629 records, 178 studies were included (112 assessed in full text, 66 on abstract values), contributing 572 ID–OOD comparisons. Retention, the ratio of OOD to ID performance, had a median of 0.82 across 133 studies with bounded metrics (95% CI 0.74–0.89), and 0.73 in full-text studies alone. Retention was lowest for laboratory-to-field acquisition shifts (0.56), sensor changes (0.55; 4 studies) and compound shifts (0.67), against 0.87 for location and 0.91 for time. By task, it ranged from 0.61 for weed recognition to 0.92 for crop mapping and 0.95 for organ detection. Among 34 studies that tested a mitigation against a source-only model, fine-tuning on labelled target data recovered a median 89% of the gap and unsupervised domain adaptation 28%. The evidence is weaker than its volume: 71% of full-text studies reported a single run, 77% did not state how models were selected, and study selection in this review was performed by one LLM-assisted screener without human verification. In-distribution accuracy overstates performance on new data by an amount that depends on the shift and the task, and evaluations should name both.

Keywords: distribution shift; domain adaptation; out-of-distribution generalisation; deep learning; precision agriculture; systematic review

Highlights

- 178 studies give 572 paired in- and out-of-distribution results for crop models
- The median model keeps 82% of its in-distribution score; 73% in full-text studies
- Laboratory-to-field, sensor and compound shifts cost most; shifts in time cost little
- Target fine-tuning recovers 89% of the gap; unsupervised adaptation 28%
- 71% of full-text studies report one run; 77% do not say how models were selected

1 Introduction

A model that recognises leaf disease, counts wheat ears or maps maize fields is trained on data from a few fields, seasons and devices, and is then used on others. Accuracy on held-out data from the training collection says little about accuracy on a new collection. The best-known case is still instructive. A convolutional network trained on PlantVillage images reached 99.35% accuracy on held-out PlantVillage images but 31.40% and 31.69% on two small sets of images of the same crop–disease classes taken from other sources (Mohanty et al., 2016). The second pair of numbers is the one that predicts field use.

The statistical problem has a name. When the joint distribution of inputs and labels differs between training and use, error measured on the training distribution is a biased estimate of error in use (Moreno-Torres et al., 2012; Quiñero-Candela et al., 2008), and bounds on error in the new domain grow with the divergence between the

two domains (Ben-David et al., 2009). Machine learning now has benchmarks built around real shifts; WILDS includes wheat-head images from different countries and acquisition sessions (Koh et al., 2020). A large controlled comparison found that, once model selection was made fair, no domain-generalisation algorithm clearly beat ordinary empirical risk minimisation (Gulrajani and Lopez-Paz, 2020). Ecology and remote sensing reached a related conclusion from the other side. Random cross-validation of spatially structured data overstates predictive skill (Roberts et al., 2017), and predictions for conditions outside the feature space of the training data carry unknown error (Meyer and Pebesma, 2021). For a biomass map of central Africa, non-spatial validation suggested that more than half of the variance was explained, while spatial validation showed almost none (Ploton et al., 2020).

Agricultural machine learning meets every kind of shift. Images move from controlled backgrounds to field clutter; models move between farms, regions and countries; seasons change crop appearance and phenology; cameras and satellites are replaced; cultivars differ. Results obtained under experimental conditions hide factors that degrade disease recognition in practice (Barbedo, 2018). Broad reviews have catalogued architectures, datasets and in-distribution accuracy (Kamilaris and Prenafeta-Boldú, 2018; Liakos et al., 2018; van Klompenburg et al., 2020) without measuring the gap. Three recent reviews address it in part. One weighs 56 studies of deep learning in farming and reports that high scores on controlled data are often given without external, temporal or cross-site validation (Amodio et al., 2026). Another organises 42 proximal and UAV imaging studies along scene, sensor, protocol and time shifts and proposes a benchmark agenda (Barbedo, 2026). A third surveys grape leaf disease detection and finds that cross-dataset transfer usually degrades results (Ghorpade et al., 2025). Judged from their abstracts, none converts paired ID and OOD results into a common effect size across tasks and modalities. Many primary studies do report the gap: for crop-type mapping across regions and years (Kluger et al., 2021; Nyborg et al., 2022b; Wang et al., 2019; Xu et al., 2020), for crop–weed segmentation across fields, crops and robots (Bosilj et al., 2019; Gogoll et al., 2020) and for disease recognition from laboratory to field (Wu et al., 2023). They report it in incompatible units and under different designs. How large the gap is, which shifts matter most and whether current remedies close it are therefore open questions.

This review asks three. How much performance is lost when crop-production models are evaluated out of distribution, and how does the loss vary with the type of shift and the task (RQ1)? When studies apply a remedy, what share of the gap is recovered, and with how many target labels (RQ2)? Are OOD evaluations designed and reported well enough for these numbers to be trusted (RQ3)?

The review makes four contributions.

- It converts 572 ID–OOD comparisons from 178 studies into one effect measure, retention, so that disease, weed, mapping and yield models can be compared on a common scale (Section 3.3).
- It ranks shift types by the performance they cost and checks the ranking inside studies that tested more than one shift, where data, task and authors are held fixed (Tables 2 and 3).
- It measures the share of the gap that each family of remedies recovers, relative to both the ID result and a target-trained reference (Table 4).
- It appraises six aspects of evaluation practice in 112 full-text studies and derives a minimum reporting set for OOD evaluations (Table 6).

2 Methods

2.1 Protocol and reporting

The protocol fixed the review questions, eligibility criteria, search strings, sentinel studies, extraction items, appraisal items and synthesis plan on 1 October 2026, before any search was run. It was not registered. The protocol and its dated deviation log are released with the review data (Data availability). Reporting follows the PRISMA 2020 statement (Page et al., 2021); the checklist is in Appendix D.

2.2 Eligibility criteria

Eligible studies trained a supervised machine-learning model, classical or deep, for one of six crop-production tasks: (a) disease, pest or abiotic-stress recognition; (b) weed–crop discrimination; (c) detection, counting or segmentation of plants, organs or fruits; (d) estimation of plant phenotypic traits; (e) crop-type or cropland mapping; (f) yield estimation or forecasting. Three conditions had to hold.

- **I1.** The study is primary and empirical.
- **I2.** It reports performance on an in-distribution (ID) test set and on at least one out-of-distribution (OOD) test set, with the same metric and label space (or a reported common subset). The OOD domain differs from training in an identifiable factor: acquisition setting (laboratory or controlled to field), location (field, site, region or country), time (season or year), sensor or platform, or biological material (cultivar, species or growth stage).
- **I3.** The OOD result of the unadapted (“source-only”) model is reported, so that the gap can be computed.

Studies were excluded if they concerned livestock, aquaculture, forestry, post-harvest processing, soil-only mapping or land cover without crop classes (E1); created the shift only by synthetic perturbation, or trained on synthetic and tested on real data (E2); used random or k-fold splits only (E3); were reviews, editorials or non-English records, or could not be retrieved and reported no ID and OOD values in the abstract (E4); or tested on a label space disjoint from training (E5). Preprints were eligible. A preprint with a peer-reviewed version was replaced by that version.

Two design variants satisfy I2. A study that evaluates the same model specification under a random split (ID) and under a domain-defined split (OOD) of the same data qualifies, whereas a single leave-one-year-out or spatial cross-validation without an ID comparator does not. When every test in a study is held out in time, as in out-of-year yield forecasting, the same-region out-of-year result serves as the ID value for a location shift. A study that reports source-only and target-trained (“oracle”) results on the OOD test set but no ID result was included under code INCLUDE-O and contributes to the oracle-referenced analyses only. Nine operational rules were fixed when a study type first appeared and then applied to all records (Table B1). The most consequential place robot-navigation perception and variety or species identification outside tasks (a)–(f), and exclude grassland and forage.

2.3 Information sources and search

We searched OpenAlex (Priem et al., 2022) and Semantic Scholar (Kinney et al., 2023) in title and abstract for publication years 2015–2026; in OpenAlex the search was restricted to articles, preprints and reviews. The final string required a crop-domain term and either a machine-learning shift term, or a learning-method term together with a broader shift phrase (Appendix A). Semantic Scholar was searched on 1 October 2026 and OpenAlex on 2 October 2026 at 00:36 UTC, after its daily request quota had reset. Licensed indexes (Scopus, Web of Science) were not available.

Search adequacy was tested against 13 reports of 12 sentinel studies named in the protocol. The sentinel studies that the search did not retrieve were sought directly (“other methods” in Fig. 1). Recall is reported in Section 3.1. Backward and forward citation chasing of the included studies, planned in the protocol, was not performed: the bibliographic services it requires were not reachable from the analysis environment.

2.4 Selection process

Records were deduplicated on DOI, then on normalised title and publication year (± 1 year). An LLM-assisted screener (Claude, Anthropic) applied the eligibility criteria to every record and recorded a decision and a task code for each inclusion. At the title and abstract stage it worked from the title and keyword-in-context excerpts of the abstract (at most two windows of ± 14 words around shift terms, at most 400 characters), read the full abstract of borderline records, and judged records without an abstract on title alone. Screening was inclusive. A record went forward when the excerpt showed evaluation on a domain other than the training domain, or when the title named domain adaptation, domain generalisation, cross-domain testing or transferability. Reports of one study were linked and the most complete report was used.

Full texts were retrieved in three passes: open-access PDFs listed by OpenAlex and Semantic Scholar; publisher article pages rendered in a headless browser, accepted only when they held a complete article; and arXiv versions matched on title. Retrieved reports were assessed against all criteria. Every exclusion was coded. Reports that could not be retrieved were assessed on their abstracts under rule E4. A study whose abstract gives the ID and OOD values of the same model was included on those values (code INCLUDE-A); all others were excluded. Eligible full-text studies whose ID and OOD values appear only in figures were recorded (code FIG) but not digitised.

Large language models used as screeners have shown chance-corrected agreement with human reviewers ranging from none to moderate, depending on the share of eligible records (Khraisha et al., 2024). No second human screener verified the decisions in this review. Section 4.5 discusses the consequences.

2.5 Data collection and data items

The same LLM-assisted reviewer extracted one row per ID–OOD comparison, from the full text or, for INCLUDE-A studies, from the abstract, into a table with fixed columns: study identifiers; task; crop; data modality; shift type and description; model; metric and its direction; ID value; source-only OOD value; mitigation (none, unsupervised domain adaptation, domain generalisation or augmentation, fine-tuning on labelled target data, large-scale or self-supervised pretraining, other); adapted OOD value; number of target labels; target-trained (oracle) value; test-set sizes; number of runs; code and data availability. Every number carries its source location. It can therefore be re-checked against the table, figure or abstract it came from. When a study reported only a mitigated model, its comparison was flagged and kept out of the gap analyses. Values computed from reported numbers, such as means over listed domains, were flagged as derived.

2.6 Appraisal of evaluation practice

No appraisal tool exists for OOD evaluations in agriculture. We adapted six items from the analysis and applicability domains of PROBAST+AI (Moons et al., 2025) and from the leakage taxonomy of Kapoor and Narayanan (2023). Each item was rated low, high or unclear concern for every study assessed in full text.

- **Q1 Domain separation.** OOD data come from fields, sites, years or devices absent from training and validation.
- **Q2 Model selection.** Hyperparameters, early stopping and checkpoints were chosen without OOD labels.
- **Q3 Test size.** The OOD test size is reported and is at least 100 units (images, fields or site-years).
- **Q4 Label consistency.** The same classes and annotation protocol are used across domains.

- **Q5 Variability.** Multiple runs or confidence intervals are reported.
- **Q6 Availability.** Code or data are public.

Abstract-only studies were not appraised. They count as unclear on every item.

2.7 Effect measures

Bounded metrics (accuracy, F1, mAP, AP50, mIoU, Dice, overall accuracy) were placed on a 0–1 scale. The primary measure is retention,

$$\rho = m_{\text{OOD}}/m_{\text{ID}},$$

with the absolute gap $\Delta = m_{\text{ID}} - m_{\text{OOD}}$ in percentage points as a secondary measure. Regression outcomes are summarised as $\Delta R^2 = R_{\text{ID}}^2 - R_{\text{OOD}}^2$ and as the ratio of OOD to ID error (RMSE or MAE). They are synthesised separately. For a mitigation, the share of the gap recovered is

$$g = \frac{m_{\text{adapted}} - m_{\text{source}}}{m_{\text{ID}} - m_{\text{source}}},$$

where m_{source} is the source-only OOD value. It equals 0 when the remedy changes nothing and 1 when it restores ID performance; values above 1 arise when the adapted model exceeds the ID reference. Because g is unstable when its denominator is near zero, it was computed only when the source-only gap exceeded 0.01 on the 0–1 scale (or 0.01 in R^2). The oracle-referenced share g_o replaces m_{ID} with the target-trained value on the same OOD test set, which holds the test data fixed. Oracle retention, $m_{\text{OOD}}/m_{\text{oracle}}$, is reported for the same reason.

2.8 Synthesis methods

Each study contributes one value per subgroup: the median of its comparisons in that subgroup. A study that reports forty comparisons therefore weighs the same as a study that reports one. We report medians with interquartile ranges (IQR) and 95% intervals for the median from a cluster bootstrap that resamples studies (10,000 draws, fixed seed) (Field and Welsh, 2007). An inverse-variance meta-analysis was not attempted. Most studies do not report the sampling variance of their metric, and the metrics differ across tasks, so heterogeneity is described rather than pooled away.

Shift type was classified as a single factor, or as compound when a comparison changed two or more factors at once; a secondary analysis counts each compound comparison under every factor it involves. Pre-specified moderators were shift type, task, data modality, model family and publication year. Data modality and model family were assigned by fixed regular-expression rules over the extracted modality and model descriptions; the rules are in the analysis code. Training-set size, also pre-specified, was not analysed. Too few studies reported it in comparable units.

Shift type and task are confounded: laboratory-to-field shifts occur almost only in disease studies, and shifts in time mostly in mapping. Two analyses address this. A task-by-shift cross-tabulation shows each shift within tasks. Within-study contrasts compare shift types, or model families, inside the same study, where data, task and authors are held fixed; the direction of the differences was tested with an exact two-sided sign test. The within-study contrasts were not in the protocol and are labelled post hoc; their p values are unadjusted, and a Bonferroni threshold for the four tests is 0.0125. Bootstrap intervals for subgroups of fewer than ten studies are reported but are unreliable.

Sensitivity analyses repeat the overall retention estimate after removing abstract-only studies, preprints, studies with high concern on Q1 or Q2, and derived values, and after restricting to studies with an OOD test set of at least 100 units and to accuracy-type metrics. The accuracy-only restriction replaces the planned chance correction, $(\text{acc} - 1/K)/(1 - 1/K)$, which needs the number of classes K for every comparison. K was not extracted.

2.9 Reporting bias and certainty of evidence

Funnel plots and regression tests for small-study effects need a variance for each effect. The included studies do not report one. Reporting bias was examined instead by comparing studies included on abstract values with those assessed in full text, and peer-reviewed studies with preprints. No formal certainty rating such as GRADE was applied, because no rating framework exists for evidence from machine-learning evaluations; the appraisal and the sensitivity analyses stand in for it.

2.10 Departures from the protocol

Table B1 lists each departure and its reason. The search string was broadened twice after sentinel tests, Semantic Scholar was added, and neither citation chasing nor the planned human re-screen was carried out. The analysis departs from the plan in four places: abstract-level inclusion under rule E4 is made explicit (INCLUDE-A), studies with values only in figures are not synthesised, training-set size is not analysed, and an accuracy-only subset replaces chance correction.

3 Results

3.1 Search and study selection

The searches returned 12,258 records: 6,646 from OpenAlex, 5,610 from Semantic Scholar and 2 returned only by an earlier iteration of the OpenAlex search (Fig. 1). After 4,629 duplicates were removed, 7,629 records were screened on title and abstract; 6,844 were excluded and 785 reports were sought. Linking 15 duplicate reports left 770 studies.

Full text was retrieved for 302 studies (39%). Of these, 151 were excluded, most often because they did not report ID and OOD values for the same model and metric (I2, $n = 109$); 40 were eligible but gave their values only in figures; and 111 were included, 10 of them with an oracle reference but no ID value. The 468 studies without full text were assessed on their abstracts: 66 reported the paired values and were included, and 385 were excluded under rule E4. Of the four sentinel studies sought directly, one could be retrieved and was included (Mohanty et al., 2016). In total, 178 studies were included.

Retrieval depended on open-access copies and was uneven across publishers. Full text was obtained for 11% of 201 Elsevier studies and 18% of 133 IEEE studies, against 68% of 130 MDPI studies and all or nearly all Frontiers and arXiv studies. The full-text set therefore over-represents open-access venues.

The search retrieved 8 of the 13 sentinel reports (8 of 12 sentinel studies). The five reports it missed (Bosilj et al., 2019; David et al., 2021, 2020; Ferentinos, 2018; Mohanty et al., 2016) do not describe their out-of-domain test in their title or abstract. Of the eight it retrieved, two were included with an oracle reference only (Nyborg et al., 2022b; Singh et al., 2020), one was included on abstract values (Xu et al., 2020) and one had values only in figures (Kluger et al., 2021). Four were excluded. Two could not be retrieved and gave no paired values in their abstracts (Gogoll et al., 2020; Wang et al., 2019), one reported no ID result (Wu et al., 2023), and one used simulated yields only (Morales and Villalobos, 2023).

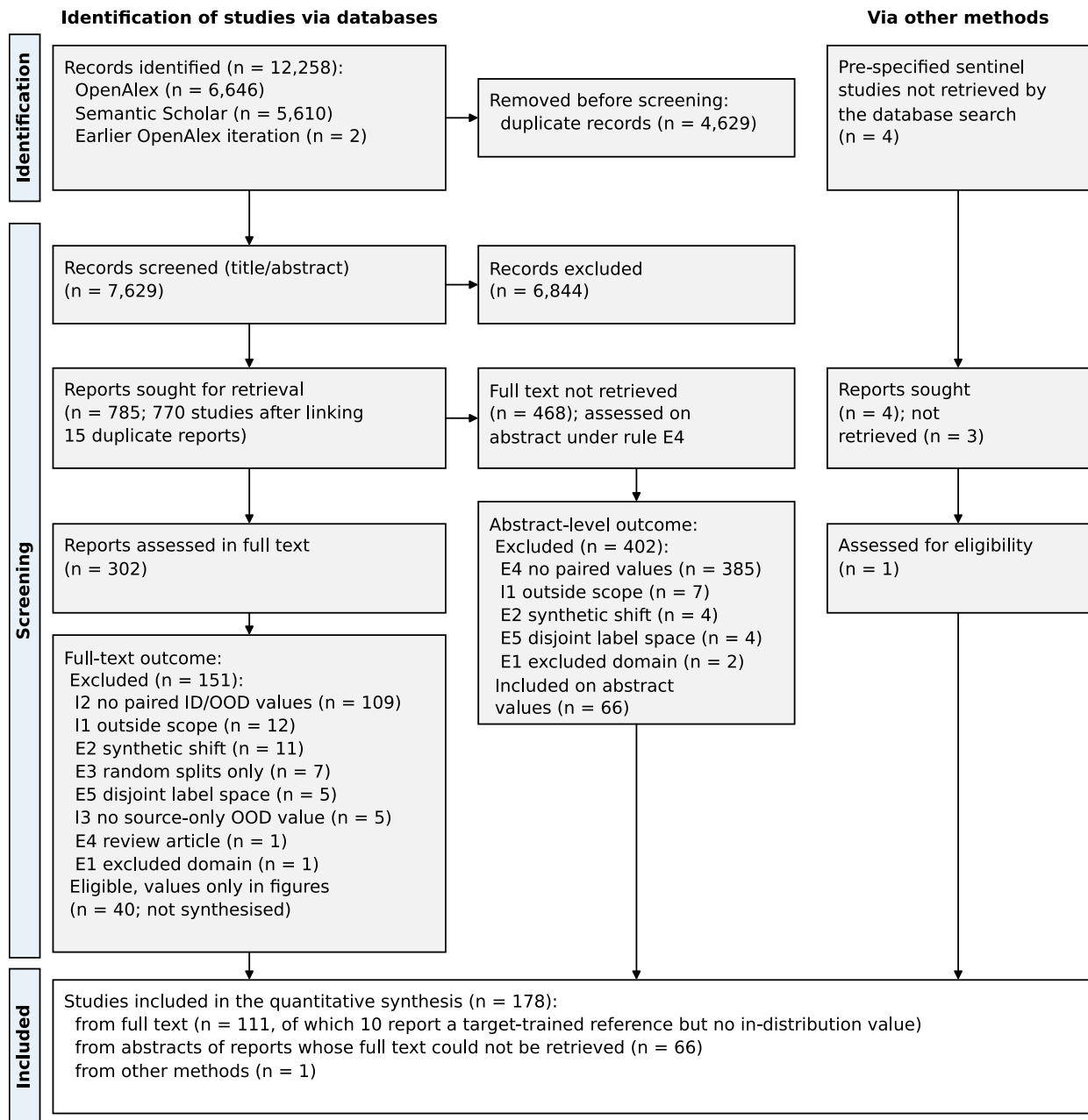


Fig. 1. PRISMA 2020 flow diagram. Studies whose full text could not be retrieved were assessed on their abstracts (middle column); rule E4 excludes those whose abstract lacks paired ID and OOD values.

3.2 Characteristics of the included studies

The evidence is recent: 85 of the 178 studies (48%) appeared in the first nine months of 2026, and 14 before 2021 (Table 1). Disease recognition (63 studies) and crop mapping (49) dominate, and proximal RGB imagery (67) and satellite data (64) are the main modalities. Location (59), compound (55) and time (54) shifts were studied far more often than shifts in sensor (5) or biological material (9). One dataset shapes the laboratory-to-field evidence. In 19 studies with a bounded metric, PlantVillage, the laboratory collection behind the opening example, was one of the two domains, and in all but one it was the training domain.

Table 1. Characteristics of the 178 included studies. A study can count under more than one task, modality or shift type, so those panels can sum to more than 178.

Characteristic	Studies, n (%)
Basis of data extraction	
Full text	112 (63)
Abstract only (full text not retrieved)	66 (37)
Publication status	
Peer-reviewed	153 (86)
Preprint	25 (14)
Publication year	
2016–2020	14 (8)
2021–2023	34 (19)
2024–2025	45 (25)
2026 (to 2 October)	85 (48)
Task	
Disease, pest and abiotic stress	63 (35)
Crop-type and cropland mapping	49 (28)
Weed–crop discrimination	22 (12)
Yield	20 (11)
Phenotypic traits	13 (7)
Plant and organ detection, counting	12 (7)
Data modality	
Proximal RGB	67 (38)
Satellite or airborne	64 (36)
UAV	32 (18)
Proximal multispectral, hyperspectral or thermal	12 (7)
Tabular or weather	5 (3)
Shift type	
Location	59 (33)
Compound (two or more factors)	55 (31)
Time (season, year)	54 (30)
Acquisition (laboratory to field, imaging setup)	27 (15)
Biological material	9 (5)
Sensor or platform	5 (3)
Outcome available for synthesis	
Bounded metric, ID and OOD (retention)	133 (75)
Regression metric only (ΔR^2 , error ratio)	29 (16)
No ID value or mitigated model only	16 (9)
Mitigation with source-only comparator	44 (25)

3.3 Size of the generalisation gap (RQ1)

Across the 133 studies with a bounded metric, the median study retained 0.82 of its ID performance (IQR 0.57–0.96; 95% CI 0.74–0.89). In absolute terms the median gap was 14.7 percentage points (IQR 3.8–37.8). The distribution is wide. 61% of studies lost more than a tenth of their ID performance, 36% more than three-tenths and 20% more than half, while 10% matched or exceeded their ID result (Fig. 3). At the extreme, a mildew detector moved from one leaf dataset to another kept 0.2% of its mAP@0.5 (Canoski et al., 2026), and a cocoa-mapping model built on pretrained satellite embeddings fell from an F1 of 91.4% to 5.8% when moved from Côte d’Ivoire and Ghana to Nigeria (Mammadov et al., 2026).

Retention varied most with the type of shift and with the task (Table 2, Fig. 2). Laboratory-to-field and other acquisition shifts retained a median 0.56 (23 studies) and sensor or platform changes 0.55 (4 studies). Compound shifts retained 0.67 (45), location shifts 0.87 (40) and shifts in time 0.91 (38). The 5 studies of a pure biological shift retained 0.93, with a wide interval (0.37–0.98). When compound comparisons are counted under each factor they involve, biological shifts fall to 0.66 (14 studies).

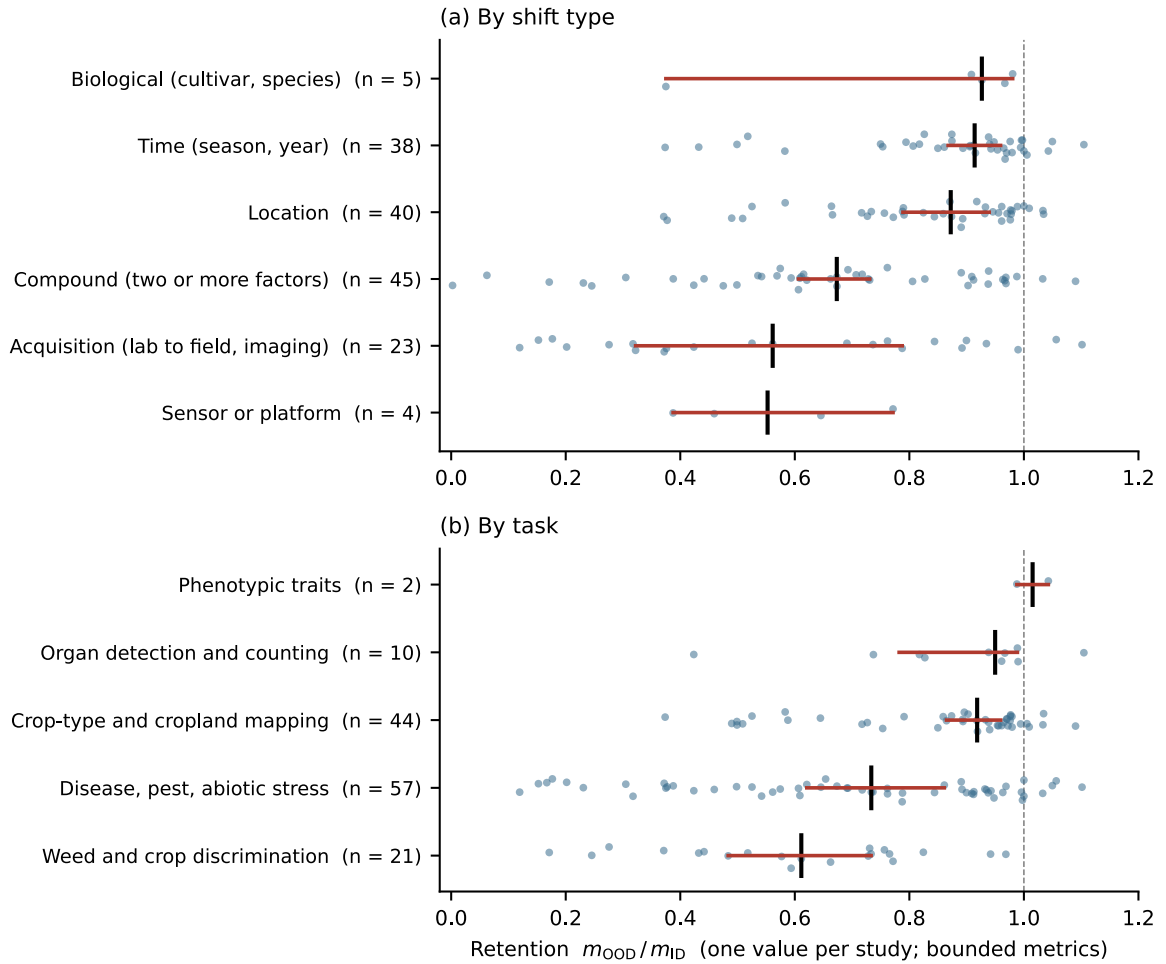


Fig. 2. Retention of in-distribution performance under distribution shift, by shift type (a) and by task (b). Each point is one study (median of its comparisons in that subgroup); black bars, subgroup medians; red bars, 95% cluster-bootstrap intervals for the median. Bounded metrics only. The dashed line marks no loss.

Table 2. Retention of in-distribution performance under distribution shift ($\rho = m_{\text{OOD}}/m_{\text{ID}}$, bounded metrics), one value per study and subgroup. IQR, interquartile range of study values; CI, cluster-bootstrap interval (10,000 resamples of studies).

Subgroup	Studies	Median retention	IQR	95% CI of median
All studies	133	0.82	0.57–0.96	0.74–0.89
<i>Shift type (primary classification)</i>				
Acquisition (laboratory to field, imaging setup)	23	0.56	0.32–0.87	0.37–0.79
Sensor or platform	4	0.55	0.44–0.68	0.39–0.77
Compound (two or more factors)	45	0.67	0.54–0.90	0.61–0.73
Location	40	0.87	0.73–0.96	0.79–0.94
Time (season, year)	38	0.91	0.82–0.97	0.87–0.96
Biological material	5	0.93	0.91–0.97	0.37–0.98
<i>Shift factor (compound shifts counted under each factor)</i>				
Acquisition	43	0.67	0.38–0.90	0.56–0.76
Sensor or platform	10	0.57	0.47–0.71	0.46–0.73
Biological material	14	0.66	0.39–0.96	0.37–0.95
Location	75	0.79	0.61–0.94	0.72–0.89
Time	50	0.91	0.72–0.97	0.83–0.94
<i>Task</i>				
Weed–crop discrimination	21	0.61	0.44–0.76	0.48–0.73
Disease, pest and abiotic stress	57	0.73	0.50–0.93	0.62–0.86
Crop-type and cropland mapping	44	0.92	0.75–0.97	0.87–0.96
Plant and organ detection, counting	10	0.95	0.82–0.98	0.78–0.99
Phenotypic traits	2	1.02	1.00–1.03	0.99–1.04
<i>Data modality</i>				
Proximal RGB	60	0.68	0.43–0.86	0.61–0.75
UAV	22	0.82	0.55–0.94	0.60–0.94
Proximal multispectral, hyperspectral or thermal	8	0.92	0.75–0.96	0.73–0.99
Satellite or airborne	45	0.93	0.75–0.98	0.87–0.97
<i>Model family</i>				
Convolutional network	85	0.73	0.50–0.92	0.62–0.77
Transformer or attention-based	41	0.76	0.55–0.91	0.66–0.88
Recurrent network	6	0.83	0.60–0.97	0.51–0.99
Other deep network (MLP, unspecified)	7	0.95	0.59–1.02	0.49–1.04
Classical machine learning	25	0.94	0.88–0.98	0.89–0.98
<i>Publication year</i>				
2016–2020	12	0.74	0.59–0.94	0.54–0.94
2021–2023	27	0.89	0.73–0.97	0.73–0.97
2024–2025	31	0.86	0.60–0.97	0.67–0.96
2026	63	0.76	0.51–0.93	0.61–0.87

Within-study contrasts point the same way. In the 9 studies that tested a compound shift and at least one single-factor shift, the compound shift retained less in 8 (median difference -0.15 ; sign test $p = 0.039$). In the 10 studies that tested both a location and a time shift, location retained less in 8 (median difference -0.04 ; $p = 0.11$), the same direction as the between-study medians. Neither contrast survives a Bonferroni correction for the four post hoc sign tests in this review, so they corroborate the ordering rather than establish it.

Tasks differ as much as shifts (Fig. 2b). Weed–crop discrimination retained 0.61 and disease recognition 0.73, against 0.92 for crop mapping and 0.95 for plant and organ detection. Part of this difference is the mix of shifts each task faces; Table 3 separates the two. Within disease recognition, acquisition shifts retained 0.56 and compound shifts 0.67, while the eight disease studies tested across seasons retained 0.97. Within crop mapping, location (0.89) and time (0.91) shifts cost about the same, and compound shifts more (0.76). Weed models retained between 0.60 and 0.76 under every shift studied three or more times. Laboratory-to-field disease studies involving PlantVillage retained a median 0.47 (14 studies), against 0.62 for those using other data (9 studies), but the intervals overlap (0.32–0.77 and 0.15–1.06).

Table 3. Median retention by task and shift type: median of study values (number of studies). Cells with fewer than three studies are shown in parentheses as descriptive only. Acq., acquisition; Sens., sensor or platform; Comp., compound; Loc., location; Biol., biological material. Yield studies used regression metrics and do not appear.

Task	Acq.	Sens.	Comp.	Loc.	Time	Biol.
Disease	0.56 (21)	(0.55; 2)	0.67 (21)	0.79 (5)	0.97 (8)	(0.65; 2)
Weed	(0.28; 1)	(0.58; 2)	0.60 (12)	0.76 (5)	0.75 (5)	(0.91; 1)
Organ	(0.99; 1)		0.63 (4)	0.96 (3)	0.83 (3)	(0.97; 2)
Mapping			0.76 (8)	0.89 (27)	0.91 (21)	
Trait			(0.99; 1)		(1.04; 1)	

Data modality follows the task pattern: satellite and airborne models retained 0.93, UAV models 0.82 and proximal RGB models 0.68 (Table 2). Across studies, classical machine-learning models retained more (0.94) than convolutional (0.73) or transformer-based models (0.76). Classical models, however, were used for satellite crop mapping in 22 of 25 studies, so this comparison mixes model with task. Inside the same study the picture changes. In 17 studies that compared both, the transformer or attention-based model retained more than the convolutional model in 14 (median difference 0.04; unadjusted $p = 0.013$). Only 5 studies compared classical and convolutional models directly, too few for a conclusion. Retention showed no monotonic trend with publication year (Table 2).

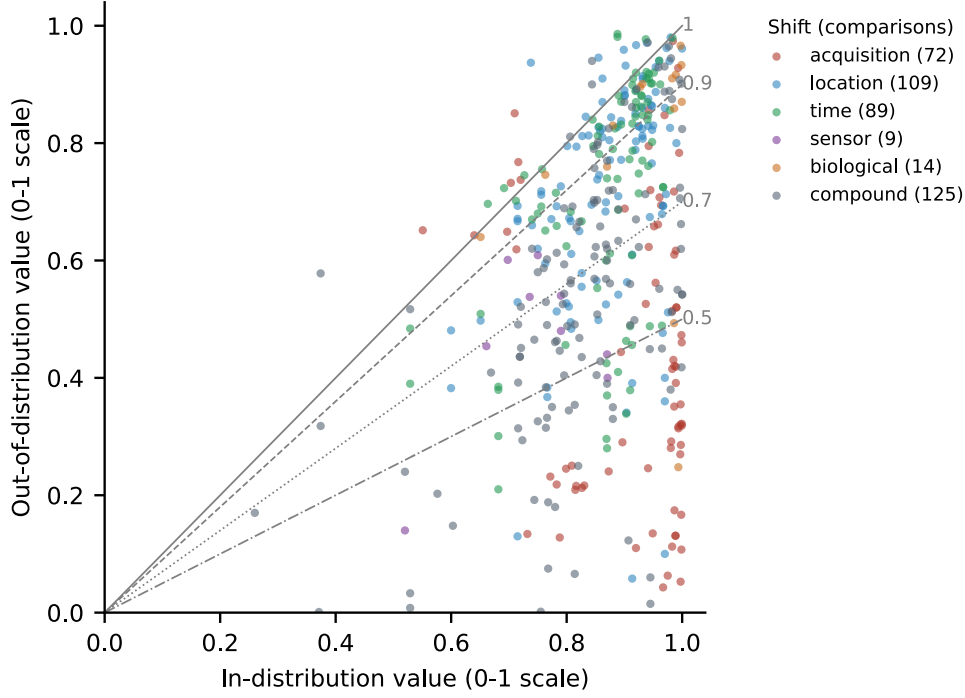


Fig. 3. In-distribution against out-of-distribution performance for 418 comparisons with bounded metrics, coloured by shift type. Grey lines mark retention of 1.0, 0.9, 0.7 and 0.5.

Regression outcomes, mostly from yield and trait models, add one twist. Across 25 studies the median loss in R^2 was 0.12 (IQR 0.05–0.25). It was larger for shifts in time (0.23, 12 studies) than in location (0.08, 10 studies), the reverse of the ordering for bounded metrics. The 3 studies that reported only an error metric showed OOD errors 1.37 times their ID errors. Against a target-trained reference on the same OOD test set, source-only models retained a median 0.74 of the oracle’s performance (18 studies; IQR 0.60–0.92), close to the ID-referenced estimate.

3.4 Recovery of the gap by mitigation (RQ2)

44 studies evaluated a mitigation against a source-only comparator; in 34 the source-only gap was large enough to compute the share recovered. Across them, the median remedy recovered 55% of the gap (IQR 29–83%; Table 4, Fig. 4). Remedies differed in kind. Fine-tuning on labelled target data recovered a median 89% (9 studies), domain generalisation and augmentation 60% (11), large-scale or self-supervised pretraining 35% (4) and unsupervised domain adaptation 28% (10). Unsupervised adaptation was also the most variable, from –19% to 91% at study level. In one study adaptation made the cross-dataset result worse (Paul, 2026). Some comparisons recovered less than a tenth of the gap, for weed models moved to new crops (Prabhakar et al., 2026; Roggiolani et al., 2025) and for a PlantVillage-to-PlantDoc disease classifier (Xiang et al., 2026).

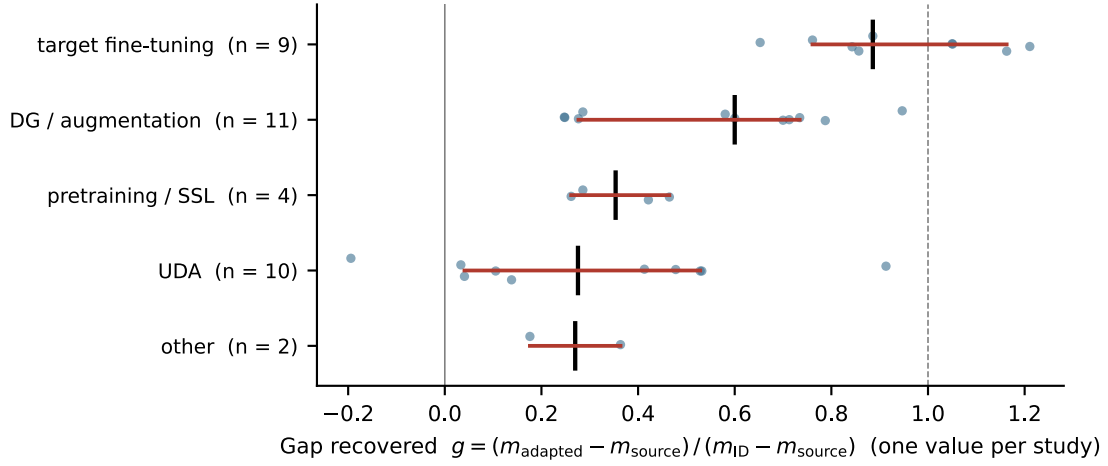


Fig. 4. Share of the out-of-distribution gap recovered, by mitigation type, one value per study. $g = 0$, no recovery; $g = 1$, in-distribution performance restored. Black bars, medians; red bars, 95% cluster-bootstrap intervals.

Table 4. Share of the out-of-distribution gap recovered by mitigation, one value per study and mitigation type. g uses the in-distribution value as the reference; g_o uses a target-trained (oracle) model on the same test set. $g = 0$: no recovery; $g = 1$: in-distribution (or oracle) performance restored.

Mitigation	Studies	Median	IQR	95% CI of median
<i>Gap recovered relative to ID, g</i>				
All mitigations	34	0.55	0.29–0.83	0.40–0.73
Fine-tuning on labelled target data	9	0.89	0.84–1.05	0.76–1.16
Domain generalisation or augmentation	11	0.60	0.28–0.72	0.28–0.73
Large-scale or self-supervised pretraining	4	0.35	0.28–0.43	0.26–0.46
Unsupervised domain adaptation	10	0.28	0.06–0.52	0.04–0.53
Other	2	0.27	0.22–0.32	0.18–0.36
<i>Gap recovered relative to oracle, g_o</i>				
All mitigations	16	0.71	0.61–0.99	0.60–0.98
Fine-tuning on labelled target data	5	1.04	1.00–1.40	0.80–1.97
Domain generalisation or augmentation	5	0.64	0.63–0.68	0.62–0.73
Unsupervised domain adaptation	6	0.48	0.40–0.87	0.23–0.98

The oracle-referenced share gives the same order. Fine-tuning reached the target-trained reference (median $g_o = 1.04$, 5 studies), domain generalisation recovered 64% of the oracle gap (5) and unsupervised adaptation 48% (6). Study-level values of g above 1 occurred only with fine-tuning. They mean that the adapted model outperformed its own ID reference, which happens when the target test set is easier than the source test set, and they are a reason to prefer g_o where an oracle exists.

The second half of RQ2, how many target labels a remedy needs, remains open. Only two studies reported the number of labelled target samples together with a source-only comparator and an ID value. Fine-tuning on 100 labelled target samples recovered 84% of the gap for a weed classifier (Milioto et al., 2017); fine-tuning on 144 recovered between –29% and 142% across four disease classifiers (Cánovas-Rodríguez et al., 2026). Two studies cannot settle the question.

3.5 Evaluation and reporting practice (RQ3)

Domain separation was rarely in doubt: 109 of 112 full-text studies (97%) tested on fields, sites, years or devices absent from training, and 98 (88%) kept classes and annotation consistent across domains (Fig. 5). The weaknesses lie elsewhere. 86 studies (77%) did not state how hyperparameters, early stopping or checkpoints were chosen, and 8 used OOD labels for selection or reported a best-of-several result. 79 (71%) reported a single run with no interval, so a difference of a few points between ID and OOD, or between a remedy and its baseline, cannot be told apart from run-to-run noise. 64 studies (57%) reported an OOD test set of at least 100 units, 9 a smaller one, and 39 did not report its size. Code or data were public for 33 studies (29%).

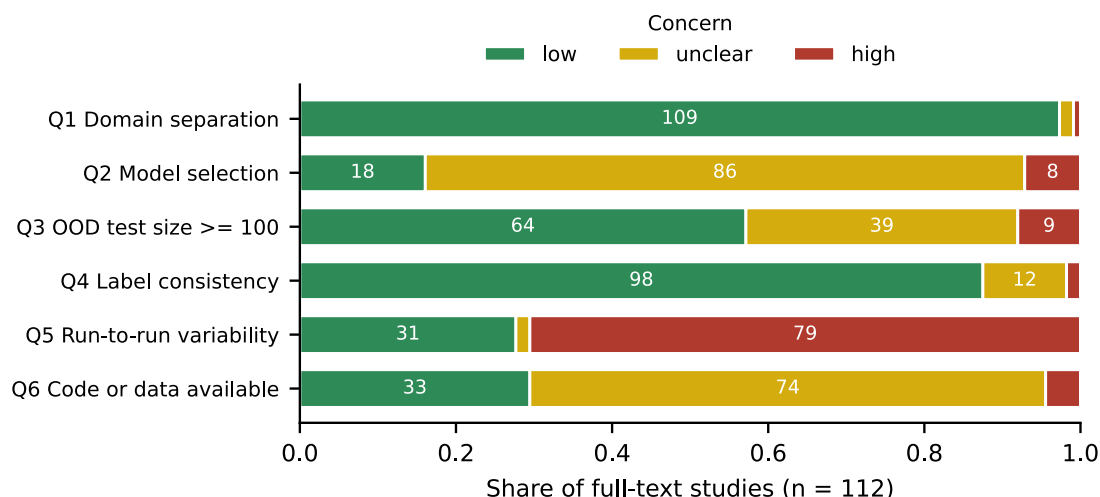


Fig. 5. Appraisal of evaluation practice in the 112 studies assessed in full text. Concern is rated low, unclear or high on each item; numbers inside the bars are study counts.

3.6 Sensitivity analyses and reporting bias

Which studies are counted matters (Table 5). Removing the abstract-only studies lowered median retention from 0.82 to 0.73, and restricting to studies with an OOD test set of at least 100 units lowered it to 0.74. Removing preprints raised it to 0.86, and the accuracy-only subset gave 0.90. Removing studies with high concern on domain separation or model selection, or removing derived values, changed the estimate by less than 0.02.

Task mix does not explain the abstract effect. Within disease recognition, abstract-only studies retained 0.86 and full-text studies 0.65; within crop mapping, 0.96 and 0.87. The ID values were similar (medians 0.91 and 0.90), so the difference lies in the OOD values that abstracts report. Two explanations fit. An abstract is more likely to quote both numbers when the transfer went well, and the model it quotes is usually the authors' proposed model rather than a plain baseline. Either way, abstract values overstate retention. The full-text estimate, 0.73, is the more conservative figure for planning.

Table 5. Sensitivity of the overall median retention to study selection.

Analysis	Studies	Median retention	IQR	95% CI of median
All studies (primary analysis)	133	0.82	0.57–0.96	0.74–0.89
Full-text extraction only (abstract-only studies removed)	84	0.73	0.51–0.92	0.66–0.80
Peer-reviewed only (preprints removed)	112	0.86	0.61–0.97	0.76–0.91
High concern on domain separation or model selection removed	129	0.82	0.58–0.96	0.74–0.89
Values read directly from the report (derived values removed)	123	0.83	0.59–0.96	0.74–0.90
OOD test set reported and ≥ 100 units	49	0.74	0.50–0.93	0.65–0.89
Accuracy-type metrics only	65	0.90	0.56–0.97	0.76–0.94

4 Discussion

4.1 Principal findings

A crop-production model evaluated on a new domain keeps, at the median, about four-fifths of its ID performance, and about three-quarters when only the better-documented full-text studies are counted. The median hides a wide spread. Some models fail almost completely; others lose nothing. Two variables go with where a model falls in that spread, the kind of shift and the task, and they are partly the same variable. Changes in how the data are captured (laboratory to field, ground camera to drone, one sensor to another) cost the most. A new location costs less, and a new season of the same fields less still. Shifts that combine factors cost more than any of their parts. The ordering holds between studies, and the within-study contrasts point the same way.

We see two mechanisms behind the ordering; both are hypotheses, not results of this review. First, acquisition and sensor shifts change the input itself: background, illumination, scale, spectral response and resolution move at once. A model trained on leaves photographed against uniform backgrounds can separate classes with cues that do not exist in field images. Second, satellite mapping models often see data processed to surface reflectance, and their features encode phenology through time series. A new year over the same region changes weather and crop calendar but not the sensor or the field structure. This would explain why mapping and satellite models retain more, and why time is the mildest shift for bounded metrics. It does not explain the regression results, where shifts in time cost more R^2 than shifts in location. Most regression outcomes come from yield models, for which a new year brings weather outside the training range. For yield, the year is the domain.

Part of the task difference is a measurement artefact. Overall accuracy in a crop map is dominated by the large classes and saturates near 1, while mIoU and mAP penalise every missed object. The accuracy-only subset retained 0.90 against 0.82 overall. Retention is therefore safer to compare within a metric family than across families, and the within-study contrasts, which hold metric, data and authors fixed, are the cleanest evidence in this review.

The transformer advantage within studies is small (median 0.04), does not survive correction for multiple testing, and carries a caveat. Where the transformer was the authors’ proposed model and the convolutional network a baseline, the comparison inherits whatever tuning advantage proposed models enjoy. We did not code which model each study proposed.

4.2 Relation to prior findings

Mohanty et al. (2016) was not an outlier. A decade later, laboratory-to-field shifts in disease recognition still retain a median 0.56, and models trained on the same laboratory collection continue to be published with ID accuracies near 100%. The finding that unsupervised domain adaptation recovers a minority of the gap matches the

controlled result of Gulrajani and Lopez-Paz (2020), where no domain-generalisation method reliably beat empirical risk minimisation once model selection was fair. In our sample 77% of full-text studies did not say how selection was done, which leaves open whether published adaptation gains survive fair selection. The cost of location shifts agrees with the ecological literature on spatial validation (Ploton et al., 2020; Roberts et al., 2017). Recent narrative reviews argued that acquisition practice and evaluation design often matter more than architecture (Barbedo, 2026) and that cross-dataset transfer degrades grape disease models (Ghorpade et al., 2025). This review puts numbers on both claims and extends them across six tasks.

4.3 Implications for research and practice

Three implications follow. First, a reported ID score should be read together with the expected retention for the shift the model will meet. On this evidence a laboratory-trained disease classifier keeps about 56% of its laboratory accuracy in the field, and a crop map transferred to another region about 89%. Second, where some labelled target data can be collected, fine-tuning is the remedy with the strongest evidence; unsupervised adaptation is a weaker substitute whose benefit varies widely between studies. Third, the evaluation itself needs to change. Table 6 lists the minimum a study should report for its OOD result to be interpretable and poolable; each item answers a gap found in this review.

Table 6. Minimum reporting set for out-of-distribution evaluations in agricultural machine learning.

Item	Report	Gap it closes in this review
1	The shift tested, named by factor (acquisition, location, time, sensor, biological) and described	Shift factors often combined and undeclared
2	ID and OOD values of the same source-only model, with the same metric and label space	109 full-text studies excluded under I2
3	Values in a table, not only in figures	40 eligible studies not synthesisable
4	How hyperparameters, early stopping and checkpoints were chosen, and on which data	77% of studies silent
5	OOD test size in its natural unit (images, fields, site-years)	39 studies did not report it
6	Several runs, or intervals, for every value	71% reported a single run
7	For a remedy: the number of labelled target samples and a target-trained reference on the same test set	Label cost reported by two studies
8	Code, model weights and the domain-level split	Public for 29% of studies

4.4 Limitations of the evidence

The included studies share four weaknesses. Most report one run, so individual retention values carry unknown noise and small differences between remedies are uninterpretable. Many do not say how models were selected, so OOD results may be optimistic. Metrics differ across tasks, which limits comparison across task families. And the literature is skewed in time and in data: nearly half the studies appeared in 2026, one laboratory dataset appears in 14 of the 21 laboratory-to-field disease studies, and only 5 studies isolated a sensor shift. Publication bias probably runs towards small gaps for proposed methods and large gaps for baselines, and the comparison of abstract-only with full-text studies shows that abstracts overstate retention.

4.5 Limitations of the review process

Six limitations concern how this review was done, and the first two matter most. First, study selection and data extraction were performed by one LLM-assisted reviewer, and no human screener verified a sample of decisions. LLM screeners can miss eligible studies (Khraisha et al., 2024), so the included set can be incomplete and extracted values can contain errors; every decision and value carries its source location in the released data, which

makes such errors checkable but not absent. Second, full text was retrieved for only 39% of candidate studies, mostly from open-access publishers, so subscription journals (including Elsevier and IEEE titles central to the field) are under-represented in the appraised set; and 66 of the 178 included studies rest on abstract values without appraisal; Section 3.6 shows that these studies report smaller gaps. Third, 40 eligible studies were not synthesised because their values appear only in figures. Fourth, citation chasing and searches of Scopus and Web of Science were not performed, and the search missed four of the twelve sentinel studies. Fifth, only English-language records were considered. Sixth, data modality and model family were assigned by regular-expression rules, which misclassify some unusual model names. The first, third, fourth and sixth can be addressed directly: a human re-screen of a random sample of decisions, digitisation of the figure-only studies, citation chasing of the included studies and a manual check of the moderator coding are the next steps for this review.

4.6 Future research

Three studies would change what is known. A benchmark that follows the same fields across years, sensors and regions, with one model family and several seeds, would separate the shift factors that this review can separate only between studies. Fine-tuning experiments that vary the number of target labels on a fixed test set would answer the label-cost question the current literature leaves open. And a controlled comparison of adaptation methods under fair model selection, as Gulrajani and Lopez-Paz (2020) did for general vision, would show which of the reported gains survive.

5 Conclusions

Across 178 studies, crop-production models evaluated out of distribution keep a median 82% of their in-distribution performance, and 73% in the full-text studies. The loss depends on the shift and the task: laboratory-to-field, sensor and compound shifts cost most, and shifts in location and time less. Fine-tuning on labelled target data closes most of the gap; unsupervised domain adaptation closes about a quarter. Most studies report one run and do not state how their model was selected. These figures are therefore approximate. An agricultural model's accuracy means little until the shift it was tested under is named.

Declarations

Declaration of generative AI and AI-assisted technologies. Claude (Anthropic) ran the database queries through scripts, screened titles, abstracts and full texts, extracted data, wrote the analysis code and drafted the text, as described in Sections 2.4 and 2.5. Every decision, extracted value, script and log is released for audit. [AUTHOR DECISION: the authors confirm that they have reviewed and edited the content and take full responsibility for it.]

Data availability. The protocol with its deviation log, search exports, deduplicated records, screening decisions, extraction and appraisal tables, analysis code and the code that builds this manuscript are in the project repository (https://github.com/2abi5/location_doner, directories `paper/review` and `paper/manuscript`). [AUTHOR DECISION: archive a release of the repository with a persistent identifier and cite it here.]

Funding. [AUTHOR DECISION]

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Appendix A. Search strings

Both databases were searched in title and abstract with the same Boolean string (Semantic Scholar syntax: + for AND, | for OR). Publication years 2015–2026; OpenAlex types article, preprint and review.

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QUERY3 = AGRI AND (SHIFT_ML OR (LEARN2 AND (SHIFT OR EXTRA)))

AGRI = (crop OR crops OR plant OR plants OR weed OR weeds OR fruit OR fruits
        OR orchard OR vineyard OR wheat OR maize OR corn OR rice OR soybean
        OR potato OR tomato OR cassava OR agriculture OR agricultural
        OR cropland OR farmland OR phenotyping)

SHIFT_ML = ("domain shift" OR "domain adaptation" OR "domain generalization"
            OR "domain generalisation" OR "out-of-distribution"
            OR "distribution shift" OR "dataset shift" OR "test-time adaptation"
            OR "unsupervised domain")

LEARN2 = ("deep learning" OR "deep-learning" OR "machine learning"
            OR "neural network" OR "neural networks" OR "convolutional"
            OR "vision transformer" OR "random forest" OR "computer vision"
            OR "semantic segmentation" OR "object detection"
            OR "image classification" OR "classifier" OR "classifiers")

SHIFT = ("domain shift" OR "domain adaptation" OR "domain generalization"
        OR "domain generalisation" OR "out-of-distribution"
        OR "distribution shift" OR "dataset shift" OR "cross-domain"
        OR "cross-dataset" OR "cross-region" OR "cross-site"
        OR "cross-season" OR "cross-year" OR "cross-location"
        OR "cross-sensor" OR "cross-crop" OR "cross-species"
        OR "unseen field" OR "unseen fields" OR "unseen region"
        OR "unseen regions" OR "unseen location" OR "unseen locations"
        OR "unseen year" OR "unseen years" OR "unseen season"
        OR "unseen environments" OR "leave-one-year-out"
        OR "leave-one-site-out" OR "leave-one-location-out"
        OR "leave-one-field-out" OR "spatial cross-validation"
        OR "spatial transferability" OR "temporal transferability"
        OR "spatiotemporal transferability" OR "spatial generalization"
        OR "temporal generalization" OR "spatiotemporal generalization"
        OR "lab-to-field" OR "laboratory to field" OR "in the wild"
        OR "test-time adaptation")

EXTRA = ("new regions" OR "new region" OR "new field conditions"
        OR "new environments" OR "different regions" OR "other regions"
        OR "different years" OR "different sites" OR "different locations"
        OR "future years" OR "transferred across" OR "spatial transfer"
        OR "temporal transfer" OR "real-world conditions" OR "non-lab"
        OR "data partitioning")

```

Appendix B. Departures from the protocol

Table B1. Logged departures from the protocol, with reasons.

Date	Departure	Reason
1 Oct 2026	Search string extended with plain-language shift phrases (EXTRA).	Sentinel recall was 2 of the first 9 resolved sentinels; the missed abstracts describe their tests in plain language.
1 Oct 2026	Semantic Scholar added as a second database.	OpenAlex holds no abstract for many Elsevier records.
1 Oct 2026	Global Wheat Head Detection dataset papers treated as a recall check only.	Dataset papers may not report paired ID and OOD results.
1 Oct 2026	Final string QUERY3 (Appendix A).	Recall 6 of 13 sentinel reports with the previous string; 8 of 13 with QUERY3.
2 Oct 2026	OpenAlex half of QUERY3 run at 00:36 UTC.	Daily request quota exhausted on 1 October.
1 Oct 2026	Operational screening rules: (a) robot-navigation perception excluded; (b) variety, cultivar and species identification excluded; (c) cropland change detection and irrigated-cropland mapping count as task (e); (d) open-set, zero-shot and cross-domain few-shot studies with disjoint label spaces are E5; (e) OOD detection without an ID-versus-OOD task metric excluded; (f) single-scheme leave-one-year-out or spatial cross-validation without an ID comparator fails I2; (g) grassland, pasture, turf and forage excluded; (h) synthetic-to-real training, including radiative-transfer simulation, is E2; (i) weather-or genotype-only yield forecasting and crop recommendation excluded.	Study types not anticipated in the protocol; each rule was fixed when first met and applied to all records.
1 Oct 2026	Title and abstract screening on title plus abstract excerpts; exclusion reasons not coded per record at this stage.	Throughput over 7,629 records; PRISMA 2020 requires reasons only at full text.
1 Oct 2026	Duplicate reports of one study linked (15 pairs).	Avoids double counting.
1 Oct 2026	Studies with a source-only and an oracle result but no ID result included (INCLUDE-O) for the oracle-referenced analyses only.	Adaptation studies often omit the source-domain result; the oracle holds the test set fixed.
2 Oct 2026	Studies without retrievable full text included on abstract values when the abstract gives the ID and OOD values of the same model (INCLUDE-A); appraisal rated unclear.	Makes the protocol’s rule E4 explicit; 468 of 770 studies could not be retrieved.
2 Oct 2026	Eligible studies with values only in figures recorded (FIG) but not digitised.	Digitisation was not feasible within the review; the studies are listed in the released data.
2 Oct 2026	Sentinel studies missed by the search sought directly (“other methods”).	Citation chasing could not be run (next row).
2 Oct 2026	Citation chasing, Scopus and Web of Science searches, and the human re-screen of a 20% sample not performed.	Bibliographic services were not reachable from the analysis environment; no second screener was available.
2 Oct 2026	Training-set size not analysed as a moderator; chance correction replaced by an accuracy-only subset; g computed only when the source-only gap exceeds 0.01; compound shifts analysed as their own category plus a factor-level analysis; modality and model family coded by regular-expression rules.	Training-set size and the number of classes were not reported consistently enough to extract; g is unstable for near-zero gaps; many comparisons changed more than one factor.
2 Oct 2026	Post hoc analyses added: task-by-shift cross-tabulation, within-study contrasts with sign tests, PlantVillage comparison, and abstract-versus-full-text comparison within task.	Shift type and task are confounded; abstract-only studies differed from full-text studies.

Appendix C. Included studies

Table C1. Studies included in the synthesis. Task: Disease (disease, pest, stress), Weed, Organ (plant and organ detection or counting), Trait, Mapping, Yield. Shift: Acq. acquisition, Loc. location, Sens. sensor or platform, Biol. biological material, Comp. compound. k , number of extracted comparisons. Effect: study median retention ρ (bounded metrics), ΔR^2 , ratio of OOD to ID error, or retention relative to an oracle ρ_o . Basis: FT full text; FT-O full text, oracle reference only; Abs abstract only.

Study	Task	Crop	Shift	k	Effect	Basis
Aarizou et al. (2026)	Disease	multiple	Acq./Biol.	18	ρ 0.50	FT
Abdalla et al. (2020)	Disease	oilseed rape	Time	1	ρ 1.00	FT
Abdoulaye et al. (2026)	Disease	multiple crops	Comp.	1	ρ 0.57	Abs
Agila and Fernandez (2026)	Disease	bean	Time	1	ρ 1.00	Abs
Alpysbay et al. (2026)	Yield	spring wheat	Loc.	1	ΔR^2 0.04	Abs
Amin et al. (2023)	Yield	winter cereals	Time	1	error $\times 1.37$	Abs
Angga et al. (2026)	Disease	chili	Comp.	1	ρ 0.89	Abs
Ardiansyah and Ilafiah (2026)	Disease	multiple crops	Acq.	1	ρ 0.12	FT
Arizo-García et al. (2026)	Trait	wheat -> barley	Comp.	1	ρ 0.99	Abs
Arsa et al. (2024)	Weed	multiple crops	Comp.	2	–	Abs
Ban et al. (2022)	Trait	rice	Comp.	6	ΔR^2 0.16	FT
Banet et al. (2024)	Trait	maize (roots)	Acq./Biol./Comp.	4	ΔR^2 0.11	FT
Bazrafkan et al. (2026)	Trait	dry bean	Comp.	1	error $\times 1.48$	Abs
Bebie and Kyparissis (2026)	Yield	durum wheat	Time	4	ΔR^2 0.42	FT
Benbahria et al. (2020)	Mapping	irrigated land (land-cover classe...	Time	2	ρ 0.37	FT
Benito-del-Valle et al. (2025)	Disease/Weed	multiple crops	Comp.	3	ρ 0.62	FT
Bohra et al. (2025)	Yield	maize	Loc.	8	–	FT-O
Boruah et al. (2026)	Disease	rice	Comp.	2	ρ 0.94	Abs
Bouacida et al. (2024)	Disease	multiple crops	Comp.	1	ρ 1.03	Abs
Bourriz et al. (2026)	Mapping	cereals	Loc.	3	ρ 0.87	FT
Canoski et al. (2026)	Disease	grapevine	Acq./Comp.	5	ρ 0.17	FT
Cánovas-Rodríguez et al. (2026)	Disease	peach	Comp.	5	ρ 0.69	FT
Cap et al. (2020)	Disease	cucumber	Loc.	1	ρ 0.73	FT
Celikkan et al. (2025)	Weed	maize	Comp.	1	ρ 0.66	FT
Chakravarty (2025)	Yield	corn; soybean	Loc.	12	–	FT-O
Chang et al. (2026)	Mapping	cropland	Loc.	1	ρ 0.58	Abs
Chen et al. (2024)	Mapping	cropland	Loc./Time	2	ρ 0.86	FT
Chiatti et al. (2023)	Organ	grapevine	Comp./Time	5	ρ 0.74	FT
Chin and Suhaili (2025)	Disease	rice	Comp.	1	ρ 0.67	Abs
Chowdhury et al. (2026)	Organ	capsicum	Acq.	1	ρ 0.99	Abs
Cornelissen et al. (2025)	Trait	grape (Brix)	Acq./Comp./Time	8	ΔR^2 0.46	FT
Deng and Lu (2026a)	Weed	lettuce and other vegetables + we...	Time	4	ρ 0.43	FT
Deng and Lu (2026b)	Weed	vegetables	Time	4	ρ 0.52	Abs
Deng et al. (2026)	Disease	wheat	Time	1	ρ 0.91	Abs

Study	Task	Crop	Shift	k	Effect	Basis
Dhawan (2026)	Disease	rice	Comp.	1	ρ 0.96	FT
Dunderski et al. (2026)	Yield	maize	Time	5	ΔR^2 0.64	FT
Ergün et al. (2026)	Disease	grapevine	Comp.	2	ρ 0.72	FT
Fan et al. (2026)	Yield	winter wheat	Comp.	1	ΔR^2 0.05	Abs
Fine et al. (2026)	Mapping	orchards (12 types)	Time	2	ρ 0.96	Abs
Gajardo et al. (2025)	Mapping	cropland	Loc.	4	ρ_o 0.93	FT-O
Gao et al. (2023)	Weed	maize + weeds	Sens.	4	ρ 0.77	FT
Gao et al. (2026)	Disease	tomato	Acq.	3	ρ 0.74	FT
Ghaffari (2021)	Mapping	cropland	Loc.	2	ρ 0.51	FT
Gholizadeh et al. (2025)	Organ	rice	Time	3	ρ 1.10	FT
Gilcher and Udelhoven (2021)	Mapping	maize	Comp./Loc./Time	12	ρ 0.64	FT
Gómez et al. (2021)	Yield	wheat	Time	1	ΔR^2 0.03	Abs
Görlich et al. (2021)	Disease	sugar beet	Time	1	ρ 1.00	FT
Gupta et al. (2023)	Disease	potato	Acq.	2	ρ 0.89	FT
Gupta and Jadon (2026)	Disease	multiple crops	Acq.	1	ρ 0.76	Abs
Guth et al. (2023)	Disease	multiple; multiple (apple, corn,...	Acq.	3	ρ 0.56	FT
Haagsma et al. (2023)	Disease	wheat	Acq.	2	ρ 0.90	Abs
Han et al. (2023)	Mapping	6 main crops	Comp.	7	ρ 0.50	FT
Han et al. (2026)	Yield	wheat	Comp.	1	ΔR^2 0.02	Abs
He et al. (2019)	Mapping	winter wheat	Loc.	5	ρ 0.93	FT
Ho-Lam and Nguyen (2026)	Disease	multiple	Acq.	4	ρ 0.20	FT
Hoppe et al. (2024)	Mapping	grass, maize, wheat, rye, rapeseed	Loc.	1	ρ 0.79	FT
Hosseini et al. (2024)	Mapping	cropping intensity classes; cropp...	Time	2	ρ 0.97	FT
Hu et al. (2025)	Organ	multiple plant species	Comp.	3	ΔR^2 0.11	FT
Ihianle et al. (2026)	Disease	strawberry	Comp.	4	ρ 0.61	FT
Ilyas et al. (2023)	Weed	bean (crop) + weeds	Comp.	8	ρ 0.73	FT
Ismail et al. (2025)	Disease	rice	Comp.	1	ρ 0.39	Abs
Jiaze et al. (2026)	Yield	maize	Loc.	1	ΔR^2 0.06	Abs
Kamangir et al. (2026)	Organ	cowpea	Biol./Loc./Time	6	ρ 0.82	FT
Kaur and Sandhu (2026)	Disease	corn, tomato	Acq.	4	ρ 0.69	FT
Khallaghi et al. (2025)	Mapping	cropland fields	Time	4	ρ 0.50	FT
Koike et al. (2024)	Trait	tomato	Loc.	1	ΔR^2 0.07	FT
Krishna et al. (2025)	Disease	multiple	Acq.	4	ρ 1.06	FT
Kumar et al. (2026)	Disease	multiple crops	Acq.	1	ρ 0.93	Abs
Kumar and Sharma (2026)	Disease	rice	Comp.	3	ρ 0.54	FT
Lauwers et al. (2024)	Weed	common bean	Loc.	3	ρ 0.37	FT
Lee et al. (2025)	Disease	rice	Acq.	2	ρ 1.10	FT

Study	Task	Crop	Shift	<i>k</i>	Effect	Basis
G. Li et al. (2023)	Mapping	maize; squash; sunflower; wheat	Time	4	ρ 1.01	FT
D. Li et al. (2023)	Mapping	cropland	Loc.	3	ρ 1.01	FT
Q.-N. Li et al. (2023)	Mapping	cropland	Time	4	ρ 0.94	FT
Li et al. (2024)	Mapping	tobacco	Loc.	3	ρ 0.98	Abs
Li et al. (2025)	Trait	rice	Time	1	ρ 1.04	Abs
F. Li et al. (2026)	Mapping	winter wheat	Time	1	ρ 0.98	Abs
L. Li et al. (2026)	Mapping	cotton	Loc./Time	2	ρ 0.92	Abs
Y.-M. Li et al. (2026)	Yield	wheat	Time	2	ΔR^2 0.12	FT
Li. et al. (2024)	Organ	Camellia oleifera	Comp.	3	ρ 0.94	Abs
Liao et al. (2023)	Disease	rice	Time	2	ρ 0.94	Abs
Lin et al. (2021)	Mapping	olive (tree crop)	Loc.	2	ρ 0.97	FT
Lin et al. (2025)	Mapping	maize (silage vs grain)	Time	2	ρ 0.94	Abs
Lin et al. (2026)	Mapping	peach	Time	1	ρ 0.95	Abs
Liu et al. (2025)	Yield	rice	Sens.	1	ΔR^2 0.06	Abs
Z. L. Liu et al. (2026)	Disease	tomato	Comp.	5	ρ 0.76	FT
S.-H. Liu et al. (2026)	Weed	multiple crops	Comp.	1	ρ 0.61	Abs
P. Liu et al. (2026)	Disease	rice	Loc.	2	ΔR^2 0.03	Abs
Lottes et al. (2018)	Weed	sugar beet	Biol./Loc./Time	5	ρ 0.73	FT
Lottes et al. (2019)	Weed	sugar beet	Comp.	2	ρ 0.44	FT
Santos Luciano et al. (2018)	Mapping	sugarcane	Time	1	ρ 0.89	Abs
Ma et al. (2021)	Yield	maize	Loc.	16	ΔR^2 0.19	FT
Ma et al. (2022)	Mapping	rice	Time	1	ρ 0.85	FT
Maleki et al. (2024)	Mapping	rapeseed	Time	4	ρ 0.75	FT
Maleki et al. (2025)	Mapping	rapeseed	Loc.	7	ρ 1.03	FT
Mammadov et al. (2026)	Mapping	cocoa	Loc.	4	ρ 0.53	FT
Martinez et al. (2021)	Mapping	multiple (tropical crops)	Time	1	ρ_o 0.72	FT-O
Martinko et al. (2026)	Organ	cauliflower; komatsuna; sugar beet	Comp.	6	ρ 0.42	FT
Melki et al. (2025)	Weed	multiple crops	Loc.	1	ρ 0.82	FT
Milioto et al. (2017)	Weed	sugar beet	Loc.	5	ρ 0.76	FT
Miranda et al. (2026)	Yield	soybean	Loc./Time	2	ΔR^2 0.15	FT
Mohammadi et al. (2023)	Mapping	corn, soybean	Loc.	5	ρ 0.98	FT
Mohanty et al. (2016)	Disease	multiple crops	Acq.	2	ρ 0.32	FT
Mohanty et al. (2026)	Disease	soybean	Sens.	1	ρ 0.46	Abs
Monzó et al. (2023)	Mapping	citrus orchards (abandonment)	Loc.	3	ρ 0.98	FT
Mooraj et al. (2026)	Disease	multiple crops	Acq.	6	ρ 0.15	FT
Naito et al. (2024)	Organ	tomato	Loc.	1	ρ 0.99	Abs
Nejadshamsi et al. (2026)	Yield	not specified (county yield)	Time	2	error $\times$ 1.29	FT
Nyborg et al. (2022b)	Mapping	multiple crop types	Loc.	5	ρ_o 0.40	FT-O

Study	Task	Crop	Shift	k	Effect	Basis
Nyborg et al. (2022a)	Mapping	multiple crops	Loc.	5	ρ 0.92	FT
Pan et al. (2021)	Disease	wheat	Loc.	1	ρ 1.00	FT
Pandžić et al. (2024)	Mapping	multiple crops	Time	1	ρ_o 0.89	Abs
Paul (2026)	Disease	rice	Comp.	3	ρ 0.61	FT
Picón et al. (2025)	Weed	multiple crops	Comp./Sens.	4	ρ 0.48	Abs
Pimpalkar (2026)	Disease	grapevine	Comp.	1	ρ 0.97	FT
Pokharel et al. (2026)	Yield	rice	Time	1	ΔR^2 0.25	FT
Prabhakar et al. (2026)	Weed	cotton; multiple crops; soybean	Comp.	7	ρ 0.17	FT
Qadri et al. (2026)	Disease	multiple crops	Acq.	1	–	Abs
Rahaman et al. (2026)	Disease	rice	Comp.	1	ρ 0.91	FT
Rahman et al. (2026)	Disease	maize	Comp.	1	ρ 0.91	Abs
Ramcharan et al. (2019)	Disease	cassava	Sens.	2	ρ 0.65	FT
Rodríguez-Vázquez et al. (2023)	Organ	pineapple	Loc.	6	–	FT-O
Roggiolani et al. (2025)	Weed	sugar beet; sugar beet -> carrot;...	Comp.	6	ρ 0.59	FT
S. and K. (2025)	Disease	rice	Comp.	1	–	Abs
Sapkota et al. (2022)	Weed	cotton -> corn; cotton -> cotton;...	Biol./Comp./Time	10	ρ 0.77	FT
Scarlat et al. (2026)	Mapping	multiple crops	Loc.	2	ρ 0.49	FT
Selea (2023)	Mapping	multiple (LPIS)	Loc./Time	4	ρ 0.73	FT
Sen and Prema (2026)	Disease	multiple crops	Acq.	1	ρ 0.84	Abs
Shao et al. (2016)	Mapping	corn, soybean	Loc.	4	ρ_o 0.97	FT-O
Shapay et al. (2026)	Disease	multiple crops	Acq.	1	ρ 0.42	Abs
Shigan et al. (2026)	Yield	green gram	Loc.	1	ΔR^2 0.41	FT
Si et al. (2026)	Disease	tomato	Time	1	ρ 1.05	Abs
Siddarth et al. (2026)	Disease	multiple crops	Acq.	2	ρ 0.53	Abs
Singh et al. (2020)	Disease	multiple crops	Acq.	3	ρ_o 0.66	FT-O
Singh et al. (2026)	Trait	wheat	Time	6	ΔR^2 0.22	FT
Singla et al. (2026)	Disease	maize	Acq.	1	ρ 0.79	Abs
Solow et al. (2026)	Trait	grapevine	Loc.	4	–	FT-O
Stiller et al. (2024)	Yield	multiple crops	Loc.	2	–	Abs
Sun et al. (2024)	Disease	peanut	Loc.	1	ρ 0.93	Abs
Sun et al. (2026)	Disease	rice	Biol.	1	–	Abs
Sykas et al. (2022)	Mapping	multiple; multiple (LPIS crop typ...	Comp./Loc.	6	ρ 0.59	FT
Tang et al. (2026)	Yield	not specified (field crops)	Loc.	1	ΔR^2 0.05	FT
Thapa et al. (2026)	Trait	winter pea (cover crop)	Time	2	ΔR^2 0.85	FT
Thirupathi et al. (2026)	Disease	multiple (field: epipremnum, mulb...	Comp.	1	ρ 0.62	FT
Triet et al. (2026)	Organ	cantaloupe	Loc.	1	ρ 0.96	Abs

Study	Task	Crop	Shift	<i>k</i>	Effect	Basis
Vegulla (2026)	Disease	multiple crops	Acq.	1	ρ 0.18	Abs
Wang et al. (2022)	Mapping	corn, sorghum, winter wheat, fall...	Comp./Loc./Time	20	ρ 0.72	FT
Wang et al. (2024)	Mapping	winter wheat	Loc.	1	ρ 0.96	Abs
Wang et al. (2025)	Mapping	soybean	Comp./Loc.	2	ρ 0.97	Abs
H. Wang et al. (2026)	Weed	barley + weeds	Loc./Time	3	ρ 0.94	FT
Z. Wang et al. (2026)	Disease	rice	Loc.	1	ρ 0.79	Abs
Wayama et al. (2024)	Disease	cucumber, eggplant, strawberry, t...	Loc.	1	ρ 0.38	FT
Weyler et al. (2023)	Weed	sugar beet + weeds	Comp.	1	ρ 0.73	FT
Wiguna et al. (2026)	Disease	tomato	Comp.	1	ρ 0.30	FT
Wijesingha et al. (2024)	Mapping	multiple crops	Comp./Loc./Time	6	ρ 0.86	Abs
Z. Wu et al. (2026)	Disease	multiple (PlantVillage)	Biol.	3	ρ 0.37	FT
J. Wu et al. (2026)	Trait	soybean	Time	2	ΔR^2 -0.04	Abs
Xiang et al. (2026)	Disease	multiple	Acq.	6	ρ 0.37	FT
Xiao et al. (2025)	Disease	maize	Time	2	ρ 0.95	Abs
Xu et al. (2020)	Mapping	corn, soybean	Loc.	1	ρ 0.96	Abs
Xu and Lu (2026)	Mapping	cotton	Comp.	2	ρ 1.09	Abs
Yadav and Congalton (2020)	Mapping	multiple crops	Loc.	3	ρ 1.03	FT
Yang et al. (2026)	Mapping	cotton, tomato, corn	Loc.	4	ρ 0.89	FT
Yi et al. (2024)	Disease	barley; winter wheat	Biol.	4	ρ_0 0.58	FT-O
Yi et al. (2025)	Disease	winter wheat, winter rye	Comp.	1	ρ 0.23	Abs
Yiyue et al. (2026)	Trait	winter wheat	Loc.	1	–	Abs
Yu et al. (2022)	Weed	wheat (lodging)	Comp.	1	ρ 0.97	FT
Zdraveska et al. (2026)	Weed	multiple (cotton, corn, vegetable...	Comp.	1	ρ 0.25	FT
Zhang et al. (2022)	Mapping	multiple crops	Time	2	ρ 0.99	Abs
Zhang et al. (2025)	Weed	rice + weeds	Acq.	10	ρ 0.28	FT
X. Zhang et al. (2026)	Organ	apple	Comp.	8	ρ 0.83	FT
S. Zhang et al. (2026)	Disease	oilseed rape	Time	1	ρ 0.86	Abs
K. Zhang et al. (2026)	Mapping	Euryale ferox (aquatic crop)	Comp.	1	ρ 0.90	FT
Jinshuai Zhang et al. (2026)	Yield	rice	Comp./Loc./Time	3	ΔR^2 0.32	FT
Jiafei Zhang et al. (2026)	Organ	multiple crops	Biol.	2	ρ 0.97	FT
Zhao et al. (2025)	Trait	foxtail millet	Loc./Time	9	ΔR^2 0.20	FT
Zhao et al. (2026)	Mapping	rice, maize, soybean	Loc./Time	24	ρ 0.90	FT
Zheng et al. (2026)	Mapping	winter wheat	Time	2	ρ 0.98	Abs

Appendix D. PRISMA 2020 checklist

Table D1. Location of each PRISMA 2020 item in this review. Item numbers follow Page et al. (2021).

Item	Topic	Location
1	Title	Title
2	Abstract	Abstract
3	Rationale	Section 1
4	Objectives	Section 1 (RQ1–RQ3)
5	Eligibility criteria	Section 2.2; Table B1
6	Information sources	Section 2.3
7	Search strategy	Appendix A
8	Selection process	Section 2.4
9	Data collection process	Sections 2.4 and 2.5
10a, 10b	Data items	Section 2.5
11	Study risk of bias assessment	Section 2.6
12	Effect measures	Section 2.7
13a–13f	Synthesis methods	Section 2.8
14	Reporting bias assessment	Section 2.9
15	Certainty assessment	Section 2.9 (not performed; reason given)
16a, 16b	Study selection	Section 3.1; Fig. 1
17	Study characteristics	Section 3.2; Table 1; Appendix C
18	Risk of bias in studies	Section 3.5; Fig. 5
19	Results of individual studies	Appendix C; released extraction table
20a–20d	Results of syntheses	Sections 3.3–3.6; Tables 2–5; Figs 2–4
21	Reporting biases	Section 3.6
22	Certainty of evidence	Sections 3.6 and 4.4
23a–23d	Discussion	Section 4
24a–24c	Registration and protocol	Section 2.1 (not registered; protocol and amendments released)
25	Support	Declarations
26	Competing interests	Declarations
27	Availability of data, code and other materials	Data availability